Problem 6

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Problem 1 *Find the derivative of the following functions:*

(a) $f(x) = \sin(x)\cos(x)$

Explanation.

$$f'(x) = (\cos(x))(\cos(x)) + (\sin(x))(-\sin(x)) = \cos^2(x) - \sin^2(x).$$

(b) $f(x) = \dfrac{e^x \tan(x)}{\sec(x) + 2}$

Explanation.

$$\begin{aligned} f'(x) &= \frac{(\sec(x)+2)(e^x\tan(x) + e^x\sec^2(x)) - e^x\tan(x)(\sec(x)\tan(x))}{(\sec(x)+2)^2} \\ &= \frac{e^x\left[(\sec(x)+2)(\tan(x)+\sec^2(x)) - \sec(x)\tan^2(x)\right]}{(\sec(x)+2)^2}. \end{aligned}$$

(c) $f(x) = e^{x\tan(x)}$

Explanation.

$$f'(x) = e^{x\tan(x)}\left(\tan(x) + x\sec^2(x)\right)$$

(d) $f(x) = \sin(x)\cos(x)e^{3x}$

Explanation.

$$\begin{aligned} f'(x) &= \frac{d}{dx}[\sin(x)\cos(x)]e^{3x} + (\sin(x)\cos(x))\frac{d}{dx}(e^{3x}) \\ &= \left(\cos^2(x) - \sin^2(x)\right)e^{3x} + 3e^{3x}\sin(x)\cos(x) \\ &= e^{3x}\left(\cos^2(x) + 3\sin(x)\cos(x) - \sin^2(x)\right). \end{aligned}$$

(e) $f(x) = \dfrac{x+5}{7x^6 + \cot(x)}$

Explanation.

$$\begin{aligned} f'(x) &= \frac{(7x^6+\cot(x))(1) - (x+5)(42x^5 - \csc^2(x))}{(7x^6+\cot(x))^2} \\ &= \frac{7x^6 + \cot(x) - (x+5)(42x^5 - \csc^2(x))}{(7x^6+\cot(x))^2}. \end{aligned}$$

(f) $f(x) = \sin(2x)\sec^3\left(x^2+4x\right)$

Explanation.

$$\begin{aligned} f'(x) &= 2\cos(2x)\sec^3\left(x^2+4x\right) + \sin(2x)3\sec^2\left(x^2+4x\right)\sec\left(x^2+4x\right)\tan\left(x^2+4x\right)(2x+4) \\ &= 2\cos(2x)\sec^3\left(x^2+4x\right) + 3\sin(2x)\sec^3\left(x^2+4x\right)\tan\left(x^2+4x\right)(2x+4)\,. \end{aligned}$$